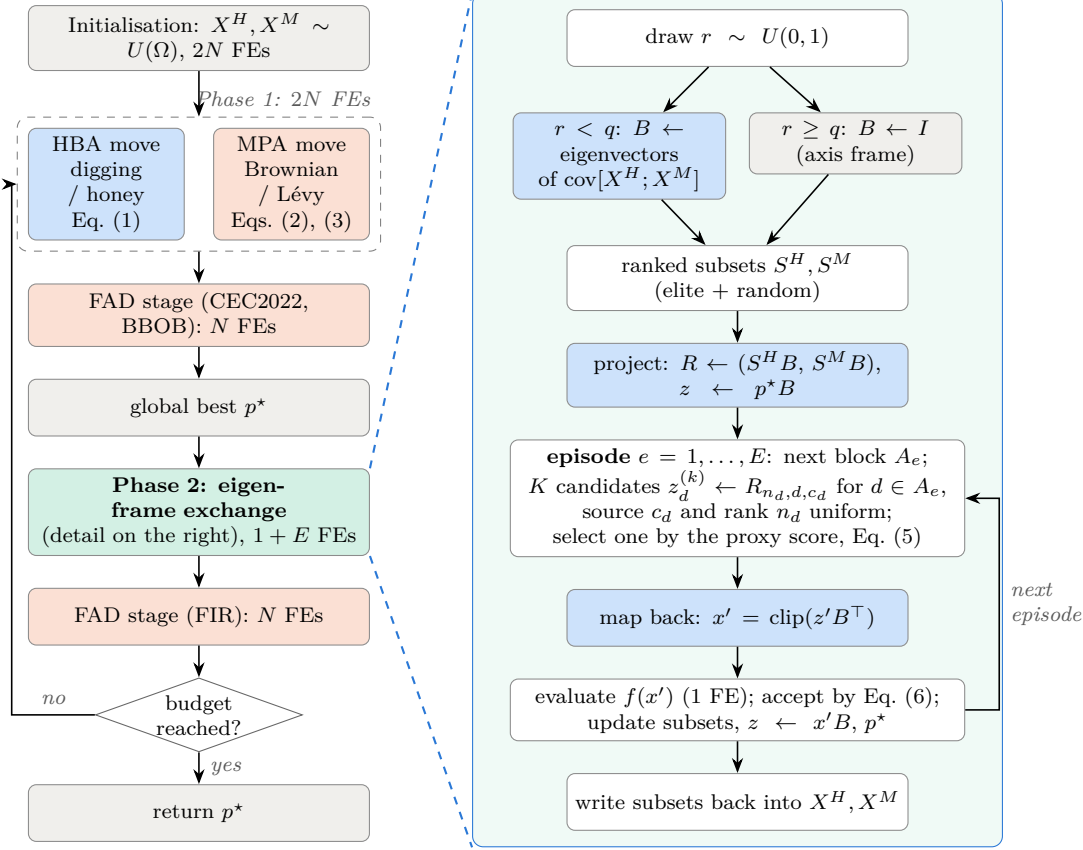

Initialisation: $X^H, X^M \sim U(\Omega)$, $2N$ FEs
Phase 1: $2N$ FEs
HBA move digging / honey Eq. (1)
MPA move Brownian / Lévy Eqs. (2), (3)
FAD stage (CEC2022, BBOB): N FEs
global best $p^\star$
Phase 2: eigen-frame exchange (detail on the right), $1 + E$ FEs
FAD stage (FIR): N FEs
budget reached?
no
yes
return $p^\star$
draw $r \sim U(0,1)$
$r < q$: $B \leftarrow$ eigenvectors of $\mathrm{cov}[X^H; X^M]$
$r \geq q$: $B \leftarrow I$ (axis frame)
ranked subsets S^H, S^M (elite + random)
project: $R \leftarrow (S^H B,\ S^M B)$, $z \leftarrow p^\star B$
episode $e = 1, \ldots, E$: next block A_e; K candidates $z_d^{(k)} \leftarrow R_{n_d,d,c_d}$ for $d \in A_e$, source c_d and rank n_d uniform; select one by the proxy score, Eq. (5)
map back: $x' = \mathrm{clip}(z' B^\top)$
evaluate $f(x')$ (1 FE); accept by Eq. (6); update subsets, $z \leftarrow x' B$, $p^\star$
write subsets back into X^H, X^M
next episode